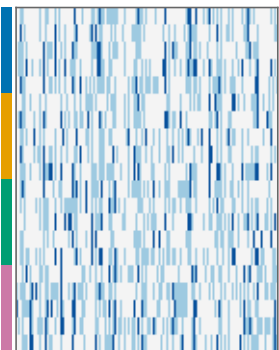


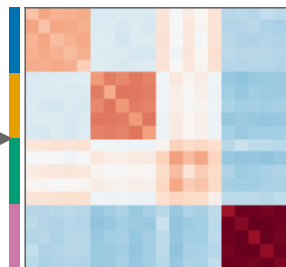
detect-white: Chen & Storey matrix, diagonal free, then whitened

genotypes G



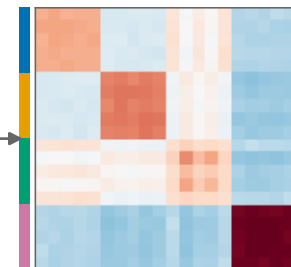
one pass

Chen & Storey H



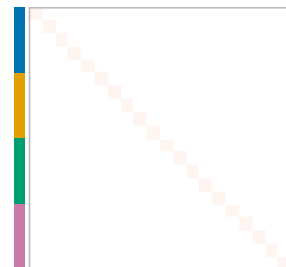
fit
 $H = L + S$

L : diag. imputed by
top 3 PCs (rank 4)



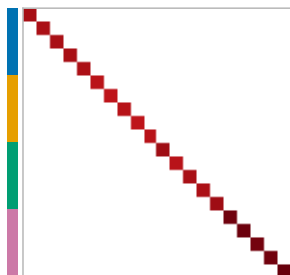
noise from the raw Gram

S : the diagonal
(small: h_i already removed)



+

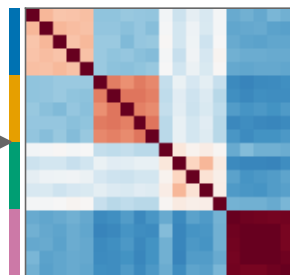
Σ : $v = \text{raw diag.} - L$
($\approx$ heterozygosity)



$\Sigma^{-1/2} A \Sigma^{-1/2}$

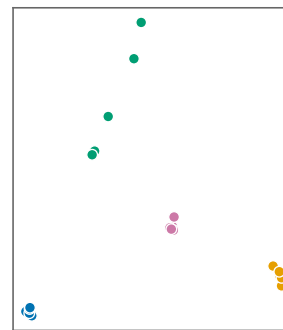
whitened:

$$A_{ij} / \sqrt{v_i v_j}$$



eigen-
vectors,
 $\times \Sigma^{1/2}$

PCs: PC2+1 vs PC3+1
MXL axis $R^2 = 1.00$



PC2+1

● CEU ● CHB ● MXL ● YRI